$$\min_{x,y} \quad F(x) + G(y) \qquad \text{s.t.} \quad h(x) + By = 0$$

$$F(x) \to f(x) + \sum_{i=1}^{s} g_i(x_i)$$

$f(x)$: Cnt Diff; $\sum_{i=1}^{s} g_i(x_i)$: LSC; $G(y)$: L-smooth (local); $h(x)$: Cnt Diff

$$\mathcal{L}_\rho(x, y, \lambda) = F(x) + G(y) + \langle h(x) + By, \lambda \rangle + \frac{\rho}{2} \| h(x) + By \|^2$$

ADMM

$$x_i^{k+1} \in \operatorname{argmin} \mathcal{L}_\rho(\underbrace{x_{\leq i}^{k+1}, x_i, x_{\geq i}^k}_{\triangleq (x_i, x_{\neq i}^k)}, y^k, \lambda^k)$$

$$y^{k+1} \in \operatorname{argmin} \mathcal{L}_\rho(x^{k+1}, y, \lambda^k)$$

$$\lambda^{k+1} = \lambda^k + \rho[h(x^{k+1}) + By^{k+1}]$$

Linearization

关于 x 全部光滑部分为 $\phi(x)$

$$\phi(x) = f(x) + \langle h(x) + By, \lambda \rangle + \frac{\rho}{2} \| h(x) + By \|^2$$

$$\frac{\rho}{2} \| h(x) \|^2 + \rho \langle h(x), By \rangle + \frac{\rho}{2} y^T B^T B y$$

$$\phi(x) \to \underbrace{\phi^k + \langle \nabla \phi^k, x - x^k \rangle + \frac{\alpha}{2} \| x - x^k \|^2}_{\text{surrogate function } u_i(x_i \mid x_i^k, x_{\neq i}^k)} \quad (\alpha > L_{\phi_i})$$

Proximal Linearized ADMM

$$x_i^{k+1} \in \operatorname{argmin}_{x_i} \langle \nabla_{x_i} \phi(x_i^k, x_{\neq i}^k), x_i \rangle + \frac{\alpha}{2} \| x_i - x_i^k \|^2 + g_i(x_i)$$

$$\nabla \phi = \nabla f + \nabla h^T \lambda^k + \rho \nabla h^T h + \rho \nabla h^T B y^k$$

$$= \operatorname{Prox}_{\frac{1}{L_{\phi_i}} g_i}\left(x_i^k - \frac{1}{L_{\phi_i}} \nabla_{x_i} \phi^k\right)$$

$$y^{k+1} \in \operatorname{argmin} \langle \nabla G^k, y \rangle + \frac{\beta}{2} \| y - y^k \|^2 + \langle h^{k+1} + By, \lambda^k \rangle + \frac{\rho}{2} \| h^{k+1} + By \|^2$$

$$= (\rho B^T B + L_G I)^{-1} [L_G y^k - \nabla G^k - B^T(\lambda^k + \rho h^{k+1})]$$

$$\lambda^{k+1} = \lambda^k + \rho[h(x^{k+1}) + By^{k+1}]$$

Convergence Analysis

1. Sufficient Descent

$$\mathcal{L}_\rho(z^{k+1}) + C \leq \mathcal{L}_\rho(z^k)$$

1) x - descent

ϕ is L_{ϕ_i}-smooth. Descent Lemma

$$\phi^{k+1} \leq \phi^k + \langle \nabla\phi^k, x_i^{k+1} - x_i^k \rangle + \frac{L_{\phi_i}}{2} \| x_i^{k+1} - x_i^k \|^2 \quad ①$$

$$x_i^{k+1} \in \operatorname{argmin} \langle \nabla_{x_i} \phi(x_i^k, x_{\neq i}^k), x_i \rangle + \frac{\alpha}{2} \| x_i - x_i^k \|^2 + g_i(x_i)$$

代入 x_i^k 得

$$\langle \nabla_{x_i} \phi^k, x_i^{k+1} \rangle + \frac{\alpha}{2} \| x_i^{k+1} - x_i^k \|^2 + g_i(x_i^{k+1}) \quad ②$$

$$\leq \langle \nabla_{x_i} \phi^k, x_i^k \rangle + g_i(x_i^k)$$

同时 $\mathcal{L}_\rho$ 关于 x 部分仅有 ϕ 和 g

故 $\mathcal{L}_\rho(x_i^{k+1}, y^k, \lambda^k) \leq \mathcal{L}_\rho(x_i^k, y^k, \lambda^k) + \frac{1}{2}(L_{\phi_i} - \alpha) \| x_i^{k+1} - x_i^k \|^2$

$$\mathcal{L}_\rho(x^{k+1}, y^k, \lambda^k) \leq \mathcal{L}_\rho(x^k, y^k, \lambda^k) - \frac{1}{2} \sum_{i=1}^{s} (\alpha - L_{\phi_i}) \| x_i^{k+1} - x_i^k \|^2$$

和 Proximal Gradient 过程一样

2) y - descent

$G(y)$ is L_G-smooth. Descent Lemma

$$G^{k+1} \leq G^k + \langle \nabla G^k, y^{k+1} - y^k \rangle + \frac{L_G}{2} \| y^{k+1} - y^k \|^2 \quad ①$$

$$y^{k+1} \in \operatorname{argmin} \langle \nabla G^k, y \rangle + \frac{\beta}{2} \| y - y^k \|^2 + \langle h^{k+1} + By, \lambda^k \rangle + \frac{\rho}{2} \| h^{k+1} + By \|^2$$

代入 y^k 得

$$\langle \nabla G^k, y^{k+1} \rangle + \frac{\beta}{2} \| \Delta y^{k+1} \|^2 + \langle h^{k+1} + By^{k+1}, \lambda^k \rangle + \frac{\rho}{2} \| h^{k+1} + By^{k+1} \|^2$$

$$\leq \langle \nabla G^k, y^k \rangle + \langle h^{k+1} + By^k, \lambda^k \rangle + \frac{\rho}{2} \| h^{k+1} + By^k \|^2 \quad ②$$

故 $\mathcal{L}_\rho(x^{k+1}, y^{k+1}, \lambda^k) \leq \mathcal{L}_\rho(x^{k+1}, y^k, \lambda^k)$

$$- \frac{1}{2}(\beta - L_G) \| y^{k+1} - y^k \|^2$$

3) λ - ascent

$$\mathcal{L}_\rho(x^{k+1}, y^{k+1}, \lambda^{k+1}) - \mathcal{L}_\rho(x^{k+1}, y^{k+1}, \lambda^k)$$

$$= \langle \underbrace{h^{k+1} + By^{k+1}}_{\frac{1}{\rho}(\lambda^{k+1} - \lambda^k)}, \lambda^{k+1} - \lambda^k \rangle = \frac{1}{\rho} \| \lambda^{k+1} - \lambda^k \|^2$$

① + ② + ③ 得

$$\mathcal{L}_\rho(z^{k+1}) \leq \mathcal{L}_\rho(z^k) - \frac{1}{2} \sum_{i=1}^{s} (\alpha - L_{\phi_i}) \| \Delta x^{k+1} \|^2 \leq 0$$

$$- \frac{1}{2} (\beta - L_G) \| \Delta y^{k+1} \|^2 \leq 0$$

$$+ \frac{1}{\rho} \| \Delta \lambda^{k+1} \|^2 \geq 0$$

Lyapunov descent $\quad \mathcal{E}^{k+1} + C \leq \mathcal{E}^k$

$$\| \lambda^{k+1} - \lambda^k \|^2 \leq \lambda_{\min}^{-1}(BB^T) \| B^T(\lambda^{k+1} - \lambda^k) \|^2$$

要用 $\Delta y^{k+1}, \Delta y^k$ 给 Bound 住

由 $y^{k+1} \in \operatorname{argmin} \langle \nabla G^{k+1}, y \rangle + \frac{\beta}{2} \| y - y^k \|^2$

$$+ \langle h^{k+1} + By, \lambda^k \rangle + \frac{\rho}{2} \| h^{k+1} + By \|^2$$

最优条件 $0 = \nabla G^{k+1} + \beta(y^{k+1} - y^k) + B^T \lambda^k + \rho B^T(h^{k+1} + By^{k+1})$

$$= \nabla G^{k+1} + \beta(y^{k+1} - y^k) + B^T \lambda^{k+1}$$

$$\| B^T(\lambda^{k+1} - \lambda^k) \|^2 = \| \nabla G^k + \beta(y^{k+1} - y^k) - \nabla G^{k-1} - \beta(y^k - y^{k-1}) \|^2$$

$$\leq \left(\| \nabla G^k - \nabla G^{k-1} \| + \beta \| \Delta y^{k+1} - \Delta y^k \| \right)^2$$

$$\leq \left(L_G \| \Delta y^k \|^2 + \beta \| \Delta y^{k+1} \| + \beta \| \Delta y^k \| \right)^2$$

$$= \left(\beta \| \Delta y^{k+1} \| + (L_G + \beta) \| \Delta y^k \| \right)^2$$

$$\leq 2\beta^2 \| \Delta y^{k+1} \|^2 + 2(L_G + \beta)^2 \| \Delta y^k \|^2$$